
Pharmacology Exam: Combined Effects of Drugs

Part I: Multiple Choice Questions (MCQs)

Select the most appropriate answer for each question.

1. General Concepts & Increased Response

1. When the combined effect of two drugs is equal to the sum of their individual effects ($1+1=2$), it is known as:
 - A. Synergism
 - B. Additive effect
 - C. Potentiation
 - D. Antagonism
2. Which of the following is a classic clinical example of an additive effect?
 - A. Levodopa and Carbidopa
 - B. Ibuprofen and Paracetamol
 - C. Activated charcoal and toxins
 - D. Adrenaline and Histamine
3. Potentiation (supra-additive) occurs when the action of one drug is enhanced by another drug that is:
 - A. More active
 - B. Equally active
 - C. Inactive
 - D. An antagonist
4. In the combination of Levodopa and Carbidopa, Carbidopa enhances the effect of Levodopa by:
 - A. Binding to the same receptor
 - B. Inhibiting the breakdown of Levodopa
 - C. Neutralizing stomach acid

- D. Acting as a competitive agonist
 - 5. Synergism is defined as a situation where the combined effect of drugs is:
 - A. Less than the sum of individual effects
 - B. Equal to the sum of individual effects
 - C. Greater than the sum of individual effects ($1+1 > 2$)
 - D. Completely abolished
 - 6. The combination of Sulphamethoxazole and Trimethoprim (Co-trimoxazole) is an example of:
 - A. Physical Antagonism
 - B. Synergism
 - C. Replacement Therapy
 - D. Chemical Antagonism
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2. Drug Antagonism

- 7. Drug Antagonism is characterized by:
 - A. An increased response
 - B. A decreased or abolished response
 - C. No change in response
 - D. Supra-additive effects
- 8. Which type of antagonism is based on the physical properties of a drug, such as the use of activated charcoal in poisoning?
 - A. Physical antagonism
 - B. Chemical antagonism
 - C. Receptor antagonism
 - D. Physiological antagonism
- 9. Antacids treating gastric acid and chelating agents treating heavy metal poisoning are examples of:

- A. Physical antagonism
- B. Chemical antagonism
- C. Competitive antagonism
- D. Synergism

10. Physiological (functional) antagonism involves two drugs that:

- A. Bind to the exact same receptor site
- B. Act at different receptors or mechanisms but produce opposite effects
- C. Form a chemical complex in the blood
- D. Shift the dose-response curve to the left

11. Which pair of drugs demonstrates physiological antagonism regarding bronchial smooth muscle?

- A. Insulin and Glucagon
- B. Acetylcholine and Atropine
- C. Histamine and Adrenaline
- D. Morphine and Naloxone

12. Adrenaline reverses bronchospasm in anaphylactic shock by acting through:

- A. Histamine receptors
- B. Adrenergic (β_2) receptors
- C. Muscarinic receptors
- D. Chemical neutralization

3. Receptor Antagonism

13. In competitive antagonism, the agonist and antagonist bind to:

- A. Different receptors
- B. Different sites on the same receptor
- C. The same site on the receptor reversibly

- D. The same site on the receptor irreversibly

14. A characteristic of the Log Dose-Response Curve (DRC) in competitive antagonism is:

- A. A flattening of the curve
- B. A rightward parallel shift
- C. A leftward parallel shift
- D. No change in the curve

15. Competitive antagonism can be overcome by:

- A. Decreasing the antagonist dose
- B. Increasing the concentration of the agonist
- C. Changing the route of administration
- D. It cannot be overcome

16. Which of the following is an example of a competitive antagonist pair?

- A. Diazepam and Bicuculline
- B. Morphine and Naloxone
- C. Insulin and Glucagon
- D. Ibuprofen and Paracetamol

17. In noncompetitive antagonism, the antagonist binds to:

- A. The same site as the agonist
- B. A different site on the receptor
- C. The agonist drug molecule itself
- D. The gastric acid

18. The Dose-Response Curve (DRC) in noncompetitive antagonism shows:

- A. A parallel shift to the right
- B. A flattening of the curve
- C. An increase in maximal effect

- D. No change in the agonist's Emax
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Part II: Essay & Short Answer Questions

Answer the following questions based on the provided material.

1. Compare and contrast Additive effect, Potentiation, and Synergism. Provide one example for each.
 2. Explain the mechanism of Physiological Antagonism and provide two examples from different physiological systems.
 3. Describe the differences between Competitive and Noncompetitive Receptor Antagonism. In your answer, include details about the binding site, how to overcome the effect, and the impact on the Dose-Response Curve.
 4. How does Adrenaline act as a physiological antagonist to Histamine during anaphylactic shock?
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Answer Key

Part I: MCQs

1. B
2. B
3. C
4. B
5. C
6. B
7. B
8. A
9. B
10. B

11. C

12. B

13. C

14. B

15. B

16. B

17. B

18. B

Part II: Essay Hints

1. Additive: $1+1=2$ (e.g., Ibuprofen + Paracetamol). Potentiation: $1+0 > 1$ (e.g., Levodopa + Carbidopa). Synergism: $1+1 > 2$ (e.g., Co-trimoxazole).
2. Physiological Antagonism: Two drugs act at different receptors/mechanisms on the same system to produce opposite effects. Examples: Insulin/Glucagon on blood sugar; Adrenaline/Histamine on bronchi.
3. Competitive: Binds same site, reversible, overcome by more agonist, rightward parallel shift of DRC. Noncompetitive: Binds different site, not overcome by more agonist, flattening of DRC.
4. Adrenaline/Histamine: Histamine causes bronchoconstriction via histamine receptors. Adrenaline causes bronchodilatation via adrenergic β_2 receptors, reversing the spasm through a different pathway.